

Pipeline Validation: POS-Tagger Migration and Associated Fixes

TODO: author name

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Abstract

This note documents a validation pass performed on the Macedonian-language stylometric pipeline after its part-of-speech tagger was migrated from spaCy (`mk_core_news_lg`) to CLASSLA-Stanza. It covers (i) how much the downstream Louvain community structure on the semantic knowledge graph (SKG) changed as a result of the tagger switch, separated from ordinary seed-to-seed algorithmic noise, including the diagnosis and subsequent fix of a community-detection artifact (author-similarity edges alone defining a 13-author community with almost no word support) and the seed-stability tradeoff that fix introduces; (ii) two correctness bugs found and fixed during the pass, one in rhyme-ending extraction and one in the style-transfer candidate-selection fallback tier; and (iii) two metric additions — a length-normalized lexical-diversity measure and stanza-level structural features — added to the per-author style profiles.

1 Background: the tagger migration

The pipeline originally tagged the corpus (2,711 Macedonian poems and song lyrics, 24 authors after the ≥ 5 -poem eligibility filter) with spaCy's `mk_core_news_lg` model. This was replaced with CLASSLA-Stanza, which lemmatizes inflection correctly and additionally exposes MULTEXT-East `xpos/feats` morphology, enabling the `feats`-keyed morphological lookup used by style transfer. The full pipeline — POS tagging, TF-IDF profiling, SKG construction and enrichment, embeddings, and graph analytics — was rerun end to end against the new tags. This note asks: how much did that rebuild actually change the pipeline's outputs, and were any of those changes symptoms of pre-existing bugs rather than the tagger switch itself?

2 Louvain community stability under the tagger switch

2.1 Setup

Author-similarity communities are computed by `graph_analytics.py` via `community.best_partition(G, weight='weight', random_state=42)` on the SKG (`skg_final.gexf`). Both the pre-switch (spaCy) and post-switch (CLASSLA) versions of `graph_analytics.py` call Louvain with these exact same parameters, so a direct diff of the two partitions is a fair comparison, not confounded by a methodology change alongside the tagger change. The pre-switch graph and analytics output were recoverable from git history (git commit `5f03983`, carried forward unchanged by the pure file-reorganization commit `363009a`); the post-switch version is the current repository state (commit `41bb49e` onward).

2.2 Direct diff against the pre-switch partition

Both partitions cover the same 24 authors and both settle on 6 communities. Comparing author community labels directly (after optimal label alignment) gives:

Metric	Value
Adjusted Rand Index (ARI)	0.221
Normalized Mutual Information (NMI)	0.483

An ARI of 1 would mean the partitions are identical (up to relabeling); an ARI near 0 means the agreement is no better than chance. 0.221 indicates a substantial reshuffling — but Louvain is itself stochastic in the order it visits nodes, so this number alone cannot distinguish “the tagger switch changed the graph structure” from “Louvain would have landed somewhere this different anyway, even on an unchanged graph, under a different run.”

2.3 Establishing a noise floor

To separate the two, Louvain was re-run 30 times on the *current* (post-switch) graph with `random_state` $\in \{0, \dots, 29\}$, holding the graph itself fixed. Each run was compared to the documented `random_state=42` reference partition:

Comparison	Min	Mean	Median
Each of 30 seeds vs. the seed = 42 reference (ARI)	0.515	0.877	0.906
All pairs among the 30 seeds (ARI, 60 sampled pairs)	0.469	0.819	—

The number of communities Louvain settles on also varies by seed alone (4, 5, or 6 communities across the 30 runs).

2.4 Conclusion: a real change, not just seed noise

Seed-only variability on a fixed graph floors at $\text{ARI} \approx 0.47\text{--}0.5$ in the worst observed case. The pre/post-switch ARI (0.221) sits *below* that floor. The tagger switch therefore produced a partition shift larger than ordinary Louvain run-to-run noise would account for — a genuine, if partial, structural change, not an artifact of re-running the algorithm once.

The change is concentrated among the “word-driven” authors (see §2.5 for the exception). Three of the six pre-switch communities (covering 14 of the 24 authors) dissolved and redistributed across different post-switch communities — e.g. the pre-switch cluster {Рацин, Поповски, Коцев, Линдро, Лазаров} split into four different post-switch communities. This is the expected effect of a tagger change: different lemmatization directly perturbs the TF-IDF and co-occurrence edges those communities are built from.

2.5 Diagnosis: Community 0, a stable but low-information clique

The current partition’s largest community (“Community 0”) groups 13 of the 24 authors around only 2 surviving word nodes:

Петре М. Андреевски, Влада Урошевиќ, Блаже Конески, Славко Јаневски, Ацо Шопов, Матеја Матевски, Богомил Ѓузел, Адем Гајтани, Веле Смилевски, Пијан Славеј, Ефтим Клетников, Ристо Лазаров, Ива Петреска

Seed stability. All $\binom{13}{2} = 78$ pairs of these authors co-clustered together in *every one* of the 30 seed runs described above — the single most reproducible feature of the entire partition, far above the general seed-noise floor.

Why it forms. Breaking down each author’s SKG weighted degree by edge type shows why: every Community-0 author draws 40–70% of its weighted degree from direct `author_similarity` edges (author ↔ author Jaccard-vocabulary-overlap edges), versus 86–100% from word-mediated (*distinctive/cooccurrence*) edges for every author outside the community. Community 0 is not a set of authors Louvain found to write alike through shared vocabulary; it is the set of authors whose word nodes did not survive strongly into the graph (consistent with the SKG’s frequency floors and node cap), leaving the direct author-similarity edges as the dominant signal available to place them by.

Not new to the tagger switch. The same phenomenon already existed pre-switch, with a 9-author core — Андреевски, Урошевиќ, Конески, Јаневски, Ѓузел, Гајтани, Славеј, Клетников, Петреска — all a subset of today’s 13, each already showing 13–47% author-similarity weight share before the tagger changed. The switch grew this core from 9 to 13 authors (adding Шопов, Матевски, Смилевски, Лазаров), it did not create it.

Interpretation. Community 0 should be read as “authors the graph has too little word-level signal to place elsewhere,” not as evidence of shared literary style — and, as described next, was subsequently removed by changing what Louvain is allowed to see.

2.6 Fix: excluding `author_similarity` edges from the partitioning graph

`graph_analytics.py` now runs Louvain on a copy of the SKG with every `author_similarity` edge removed, keeping *cooccurrence*, *semantic_similarity*, and *distinctive* edges — i.e. only edges mediated by an actual word — as the partitioning signal. `author_similarity` edges are still computed and still present in `skg_final.gexf` for every other purpose (PageRank, weighted degree, visualization); they are excluded only from the graph Louvain sees, since §2.5 showed they were sufficient on their own to define a community with no word-level support behind it. The direct author-similarity signal remains available where it was always meant to be used as an auxiliary/fallback source — in `candidate_selection.author_similarity()`, which already computed it independently of the graph, for `tier_cross_author_similar`.

Result. Community 0 no longer exists. Louvain on the restricted graph settles on 5 communities instead of 6, and every one now has substantial word support (49–102 word nodes each, versus the old Community 0’s 2). The 13 authors previously grouped by author-similarity edges alone are now distributed across all 5 communities on genuine vocabulary evidence:

Former Community-0 author	New community (by word evidence)
Блаже Конески, Славко Јаневски, Ацо Шопов	Community 2 (with Кочо Рацин)
Влада Урошевиќ, Пијан Славеј, Ефтим Клетников	Community 1
Петре М. Андреевски, Богомил Ѓузел, Ристо Лазаров, Ива Петреска	Community 3
Адем Гајтани, Веле Смилевски	Community 4
Матеја Матевски	Community 0 (new, unrelated to the old one)

Cost: the new partition is less seed-stable. Repeating the 30-seed sweep from §2 on the restricted graph gives a markedly lower stability floor than the full graph had:

Graph	Min ARI	Mean ARI	Median ARI
Full graph (incl. <code>author_similarity</code>), §2	0.515	0.877	0.906
Restricted graph (word-mediated only)	0.300	0.581	0.541

This is the expected price of the fix, not a sign it is wrong: the author-similarity edges were direct, high-weight `author↔author` connections that gave Louvain unambiguous structure to lock onto: precisely the “glue” that made Community 0 so artificially stable (§2.5). Without it, the remaining word-mediated graph is sparser, and Louvain’s well-known sensitivity to node-visit order shows up more. Per-community confidence is uneven: averaging pairwise co-clustering across the 30 seeds by the seed = 42 community gives 0.88 for Community 4 and 0.74 for Community 3 (high confidence) down to 0.49 for Community 2 and 0.27 for the 3-author Community 0 (Матевски, Линдро, Цветкоски — low confidence, on par with a coin flip for some pairs).

Recommendation. Treat the 5-community, word-mediated partition as the current best estimate of stylistic community structure, but do not read the exact `random_state = 42` assignment as ground truth for the weaker communities (2 and the new 0) the way one might have been tempted to for the old, artifactually-stable Community 0. A seed-ensemble/consensus partition would give a more defensible per-pair confidence than any single seed and is a natural next step if community membership starts being used for a downstream decision (e.g. cluster-based candidate pooling) rather than reported descriptively.

3 Correctness fixes found during validation

Two bugs, unrelated to the tagger switch, were found and fixed while validating pipeline outputs.

3.1 Rhyme-ending extraction

`style_profiler.py`’s `top_rhyme_endings` field took the literal last three characters of each line’s last whitespace-split token, with no punctuation stripping. Macedonian poetry in this corpus frequently ends lines on a standalone `em/en` dash (a caesura marker) or on punctuation glued to the last word, so every one of the 24 authors’ top-5 rhyme endings were contaminated with punctuation artifacts — e.g. Блаже Конески’s were `--|на, |на. |ам. |ат.` and Кочо Рацин’s were `--|---|ата|ите|ни, .` This was corpus-wide, not an isolated case.

The fix extracts a genuine line-final syllable: it walks backward past trailing pure-punctuation tokens, strips edge punctuation from the real last word, and returns the substring from the onset of the final vowel’s syllable (the character after the second-to-last vowel) through the end of the word. After the fix, no author’s top rhyme endings contain punctuation, and the endings reflect real Macedonian grammatical suffixes (`-та, -те, -на, -ва`, and similar).

3.2 Style-transfer legacy-fallback tier

`candidate_selection.py`’s last-resort candidate tier (`tier_legacy_fallback`) imported a hardcoded `TARGET_AUTHOR` constant from `llm_probe.py` instead of using the `target_author` parameter it was already given, so whenever this tier fired for any author other than that hardcoded default, it silently supplied *that other author*’s TF-IDF vocabulary as replacement candidates. The fix threads `target_author` through to `load_target_words()`, which now accepts an explicit author rather than reading the module constant.

To measure real-world impact, `style_transfer.transfer_style()` was re-run over 960 sampled (source poem, target author) pairs spanning all 24 authors, instrumented to record which tier produced each of the 20,940 replacement slots evaluated:

Tier of the candidate chosen in the output	Share of slots
primary	94.20%
relaxed_semantic	5.73%
legacy_fallback	0.062% (13 slots)
cross_author_similar	0.014%

The bug was real wherever it fired, but it fired rarely — 13 of 20,940 slots, concentrated in a handful of sparse-vocabulary authors (8 of the 13 for a single author, Јосип Коцев). It was corrupting a rare edge case, not a meaningful share of style-transfer output.

4 Metric additions to the author style profiles

4.1 Length-normalized lexical diversity

Plain type-token ratio (TTR) in `author_style_profiles.csv` is not length-normalized, and per-author corpus size varies 37-fold across the 24 authors (232 to 8,538 content tokens). On this corpus, plain TTR correlates with token count at Pearson $r = -0.76$ ($p = 1.7 \times 10^{-5}$) — the expected artifact of TTR mechanically falling as a corpus grows, rather than a signal of authorial lexical richness. A Moving-Average Type-Token Ratio (MATTR; Covington and McFall, 2010) column was added, averaging the TTR of every 200-token sliding window over each author’s lemma sequence (window size chosen to fit inside the smallest eligible author’s 232-token corpus). MATTR against token count: $r = -0.037$ ($p = 0.86$) — no residual length dependence. The original `type_token_ratio` column is retained for reference but is now documented as diagnostic-only; `mattr` is the column to use for cross-author comparison.

4.2 Stanza-level structure

`author_style_profiles.csv` previously had no stanza-level features. Poem text in `stripped_songs.csv` uses blank-line-delimited stanzas (the same convention already relied on elsewhere in the pipeline for stanza extraction). Two fields were added, computed per author from that delimiting: `avg_stanzas_per_poem` and `avg_lines_per_stanza`. Values are plausible and vary meaningfully across authors — e.g. Ристо Лазаров averages 7.23 stanzas/poem at 2.95 lines/stanza (short, couplet-like stanzas), while Лилјана Дирјан averages 1.14 stanzas/poem at 15.62 lines/stanza (long, largely unbroken poems).

5 Worked example: style transfer after the fixes

As a concrete illustration of where the pipeline stands after the fixes in this note, this section walks through one full style-transfer run on the current (post-fix) pipeline state: Кочо Рацин’s “Балада за непознатиот” (“Ballad of the Unknown [Soldier]”), rewritten in Блаже Конески’s style via `style_transfer.transfer_style(text, `Блаже Конески`)`. This is a real, reproducible pipeline run, not a hand-picked or edited example — it was independently regenerated against the current corpus and model files and diffs byte-identical (modulo typographic quote-mark style) to the run quoted below.

Натаму - в поле битолско
 чемрее врба проклета -

под врбата незнаен гроб,
 в гроб лежи војник непознат.
 Лежи од војна света,
 лежи — и веќе земјосал —
 силно го тага гласи
 мрзне тука загинал.
 Никој крај него немаше —
 вишното небе врз него,
 земјата скришна под него,
 над гробот врба спремам.
 (...)

И од ним — самовилите
 една по една идеа —
 од гроб до гроб го мрзне
 јунак до јунак — на оро.
 (...)

(Full output omitted here for space; see the corpus entry for the source text and re-run `style_transfer.transfer_s` for the complete transferred poem.)

What this demonstrates, and what it doesn’t. This call did not pass `source_author`, so only replacement criterion (b) — a lemma absent from the target’s vocabulary for a POS in `B_ELIGIBLE_POS = {ADJ, ADV, VERB}` — is active; criterion (a) (source-distinctive/target-absent, any content POS) needs a source author and did not fire here. This is visible directly in the output: NOUN tokens like `гроб` (‘grave’) and `гrobiшта` (‘graveyard’) are preserved verbatim throughout, exactly as `style_transfer.py`’s own design comment (labelled “Criterion (b)”) intends — nouns carry subject matter, not style, and are deliberately left alone under this criterion. Adjectives and verbs are substituted where Конески’s own vocabulary lacks the Рацин original: `војна световна` (‘worldly/mundane war’) becomes `војна света` (‘holy war’), and `тага изела` (‘grief consumed [him]’) becomes `тага гласи` (‘grief speaks/sounds’), each drawing a real Конески-attested word of the same POS into the same grammatical slot.

Two honest limitations visible in this same output. First, `задека` (‘because’) — a conjunction — is itself replaced (with `мрзне`, ‘freezes’), which looks wrong until you check the tagging: CLASSLA tags this token VERB in this context (Aspect=Perf|Tense=Past), not CONJ/SCONJ, so it passes through `CONTENT_POS` and becomes a legitimate replacement target under the (mistaken) tag. This is the same category of tagging error `pos_flags.csv`’s `upos_xpos_clash/unstable_tag` reasons are built to catch for common cases — `задека` was not one already in `data/pos_corrections.csv` at the time of this run, so it slipped through uncorrected. Second, `дигаа` (‘they raised’, Number=Plur, 3rd-person imperfect) is replaced by `мрзне`, which surfaces in an unmarked/singular-present-looking form — the morphological lookup evidently had no attested Конески plural-imperfect form of that lemma to realise into, and fell back to a lower morph tier (`nearest_attested` or `unresolved`) rather than the requested agreement. Neither of these breaks the pipeline, and neither is new to this session’s fixes — they are exactly the kind of residual, low-frequency noise the tier/morph-tier logging in `candidate_selection.py` exists to make visible and auditable, rather than silent.

6 Reproducibility

Figures in §2–§2.5 (the diagnosis) were produced against `skg_final.gexf` and `graph_analytics.csv` as of commit 41bb49e (or later, up to the fix below, on branch `feats-morph-lookup`), and against the pre-switch versions recoverable at commit 5f03983/363009a. Figures in §2.6 (the fix) were produced against `src/graph_analytics.py`'s current working-tree state on that same branch, which restricts Louvain's input graph to word-mediated edges; commit this change alongside the CSV it produces to keep the two in sync. The Louvain seed-stability sweeps and the tier-frequency measurements were produced by one-off scripts run against the live pipeline rather than checked-in tests, since they characterize algorithmic stability and empirical tier frequency rather than fixed expected behavior.